
EEM 561 Machine Vision

Lecture 2

Perspective

Spring 2025

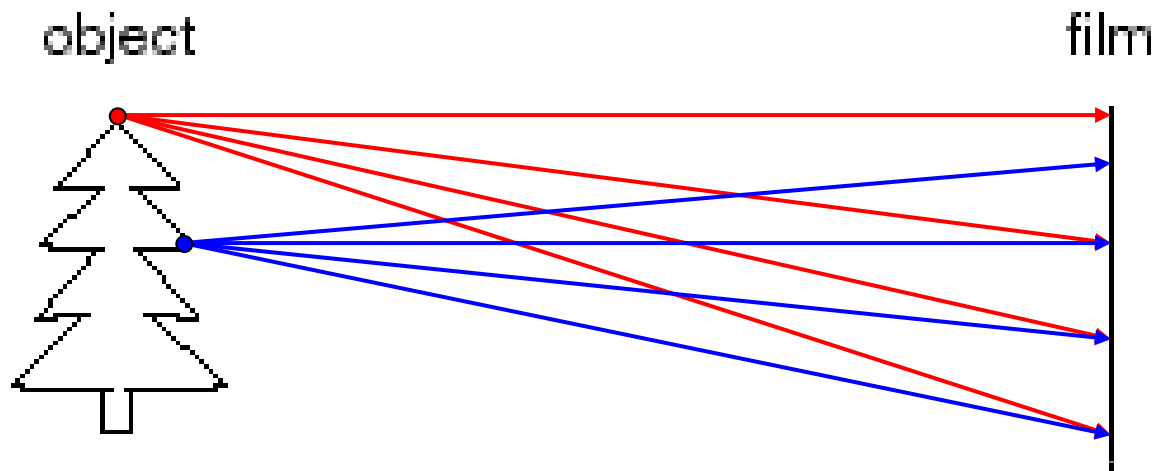
Asst. Prof. Dr. Onur Kılınç

Eskisehir Tech. Uni., Dept. EEE

Overview of next two lectures

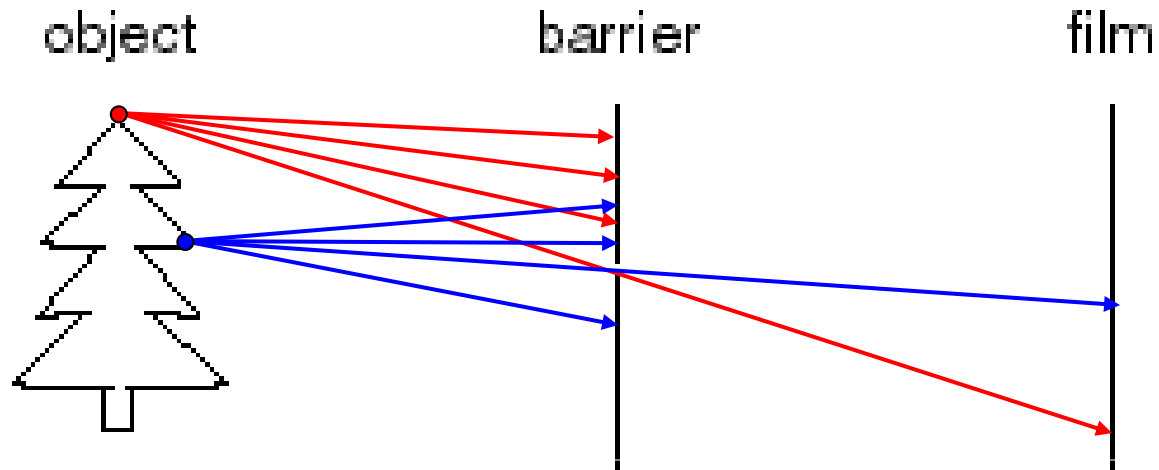
- The pinhole projection model
 - Qualitative properties
 - Perspective projection matrix
- Cameras with lenses
 - Depth of focus
 - Field of view
 - Lens aberrations
- Digital sensors

Let's design a camera



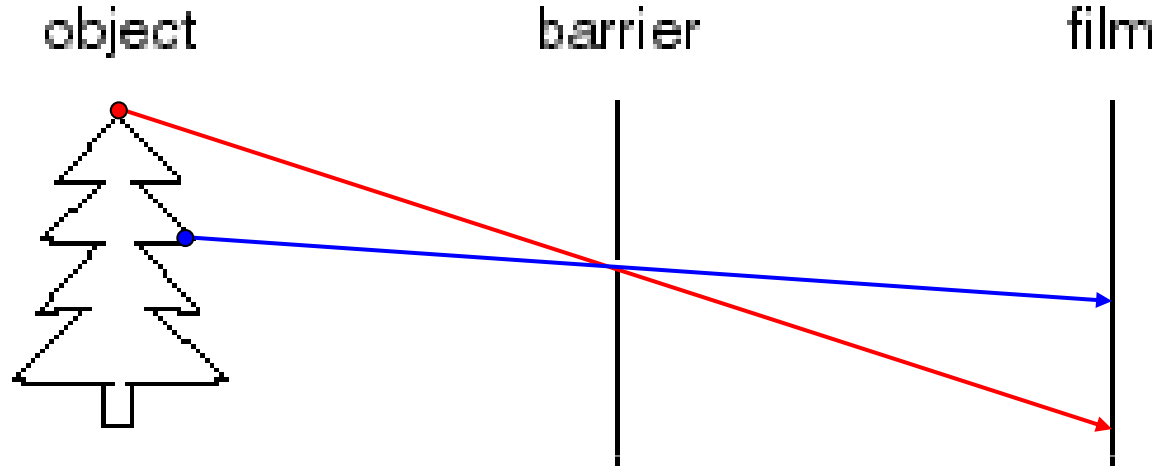
Idea 1: put a piece of film in front of an object
Do we get a reasonable image?

Pinhole camera



Add a barrier to block off most of the rays

Pinhole camera



- Captures **pencil of rays** – all rays through a single point: **aperture, center of projection, optical center, focal point, camera center**
- The image is formed on the **image plane**

Pinhole cameras are everywhere

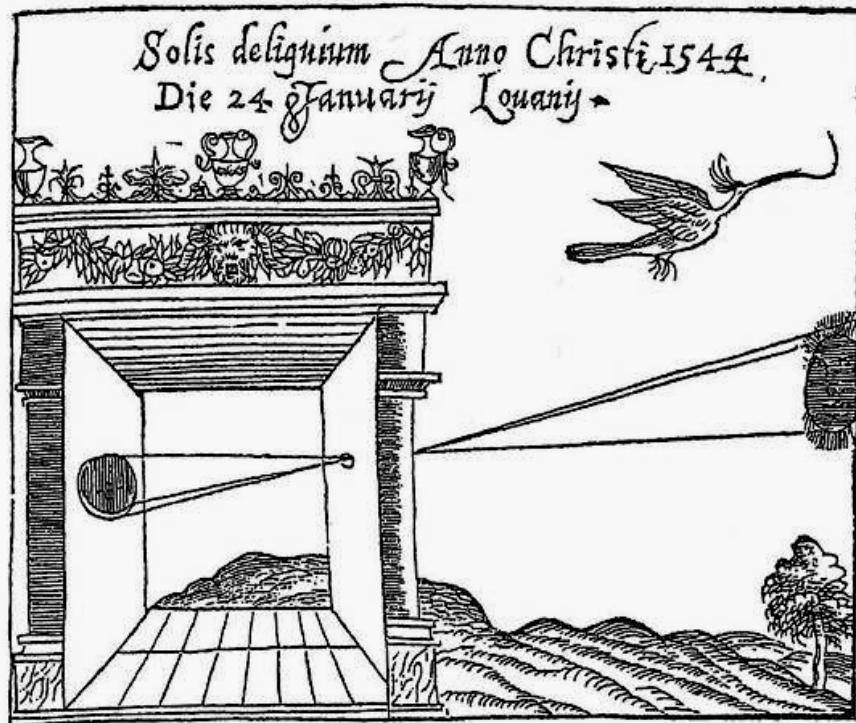


Tree shadow during a solar eclipse

photo credit: Nils van der Burg

<http://www.physicstogo.org/index.cfm>

Camera obscura



Gemma Frisius, 1558

- Basic principle known to Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)

Turning a room into a camera obscura



After scouting rooms and reserving one for at least a day, Morell masks the windows except for the aperture. He controls three elements: the size of the hole, with a smaller one yielding a sharper but dimmer image; the length of the exposure, usually eight hours; and the distance from the hole to the surface on which the outside image falls and which he will photograph. He used 4 x 5 and 8 x 10 view cameras and lenses ranging from 75 to 150 mm.

After he's done inside, it gets harder. "I leave the room and I am constantly checking the weather, I'm hoping the maid reads my note not to come in, I'm worrying that the sun will hit the plastic masking and it will fall down, or that I didn't trigger the lens."

From *Grand Images Through a Tiny Opening*, **Photo District News**, February 2005

Abelardo Morell, Camera Obscura Image of
Manhattan View Looking South in Large
Room, 1996

<http://www.abelardomorell.net/project/camera-obscura/>

Turning a room into a camera obscura

My hotel room,
contrast enhanced.

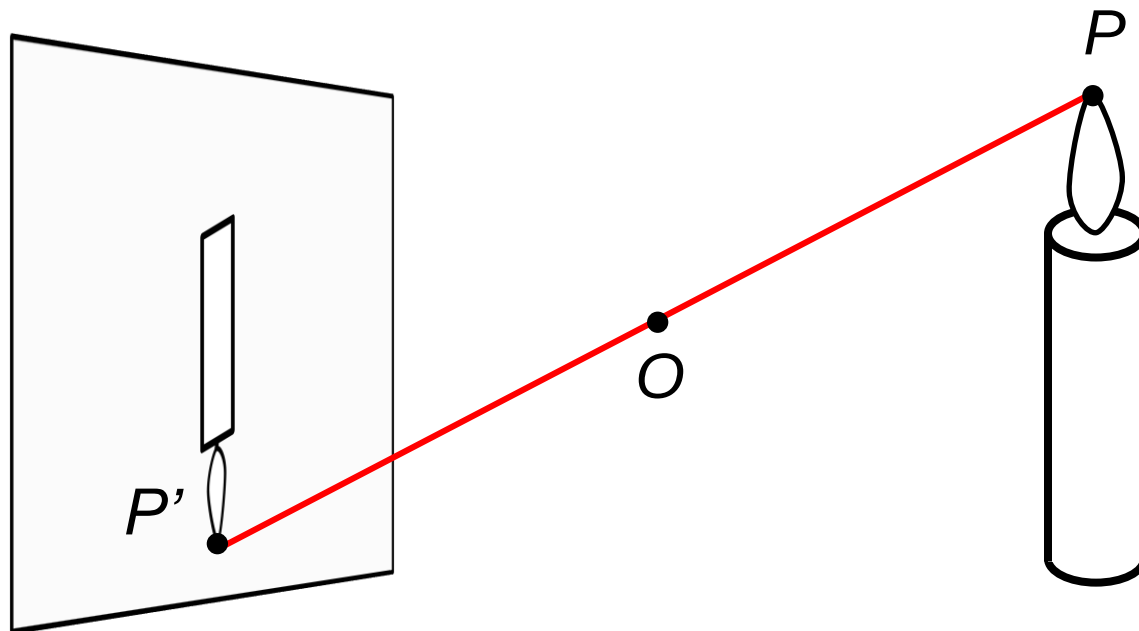


The view from my window



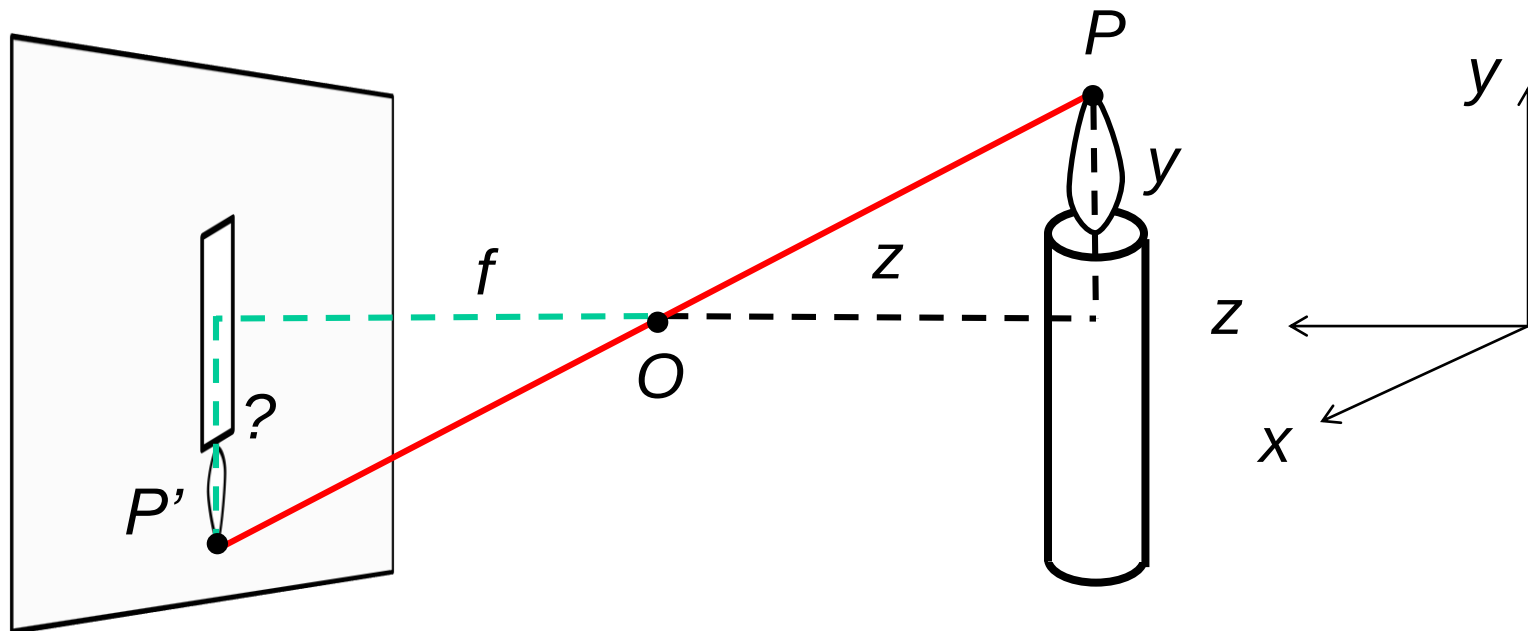
Accidental pinholes produce images that are
unnoticed or misinterpreted as shadows

Modeling projection



- How do we find the projection P' of a scene point P ?
 - Form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
- All scene points that lie on this visual ray have the same projection in the image
- Are there scene points for which this projection is undefined?

Modeling projection



The coordinate system

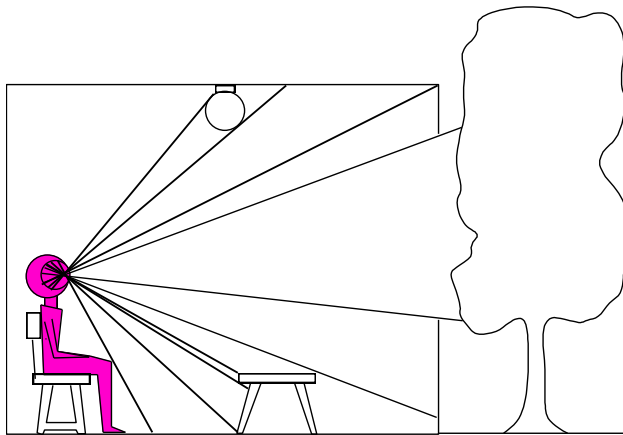
- The optical center (O) is at the origin
- The image plane is parallel to xy -plane or perpendicular to the z -axis, which is the *optical axis*

Projection equations

- Derived using similar triangles $(x, y, z) \rightarrow (f \frac{x}{z}, f \frac{y}{z})$

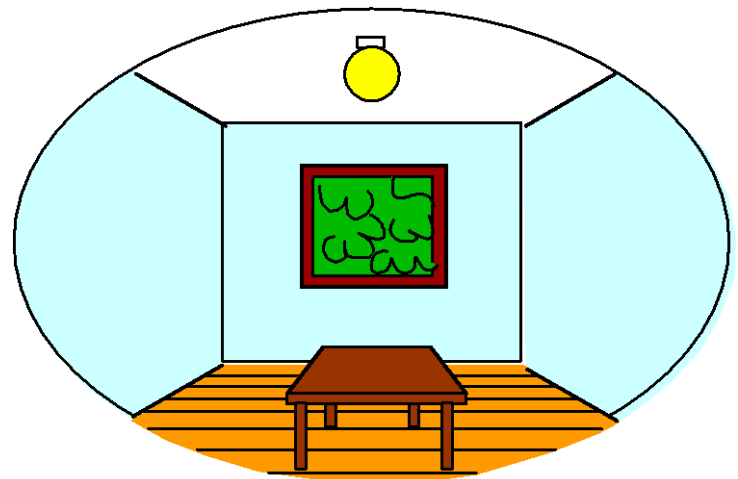
Dimensionality reduction: from 3D to 2D

3D world



Point of observation

2D image



What properties of the world are preserved?

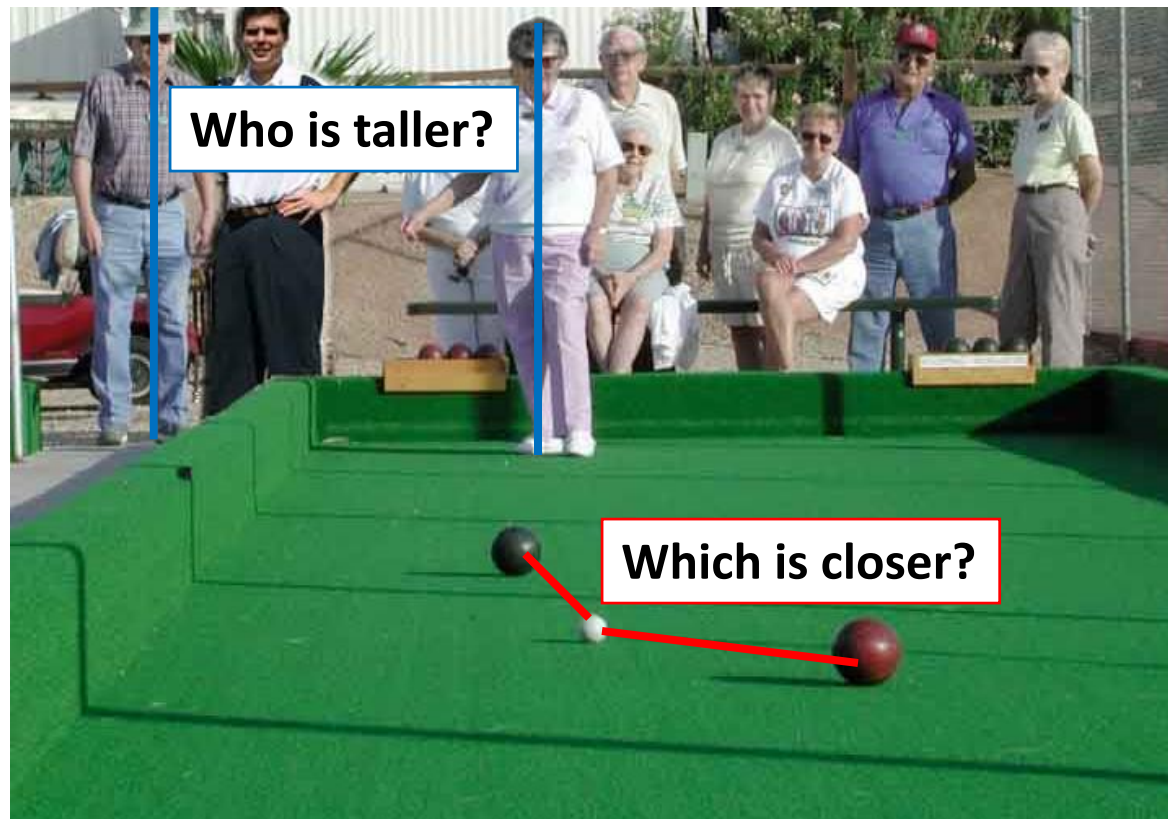
- Straight lines, incidence

What properties are not preserved?

- Angles, lengths

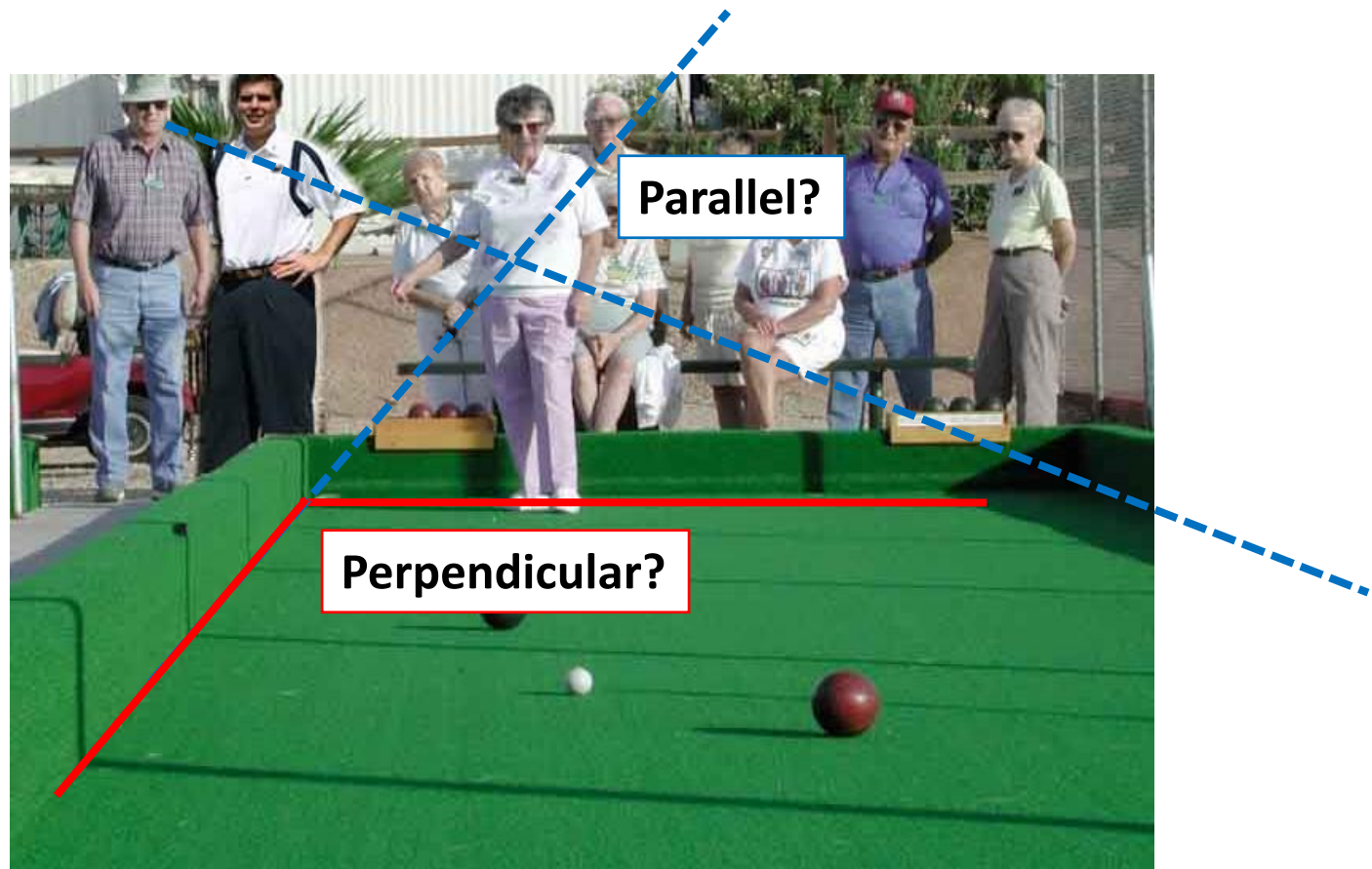
Properties of projection

What is lost?



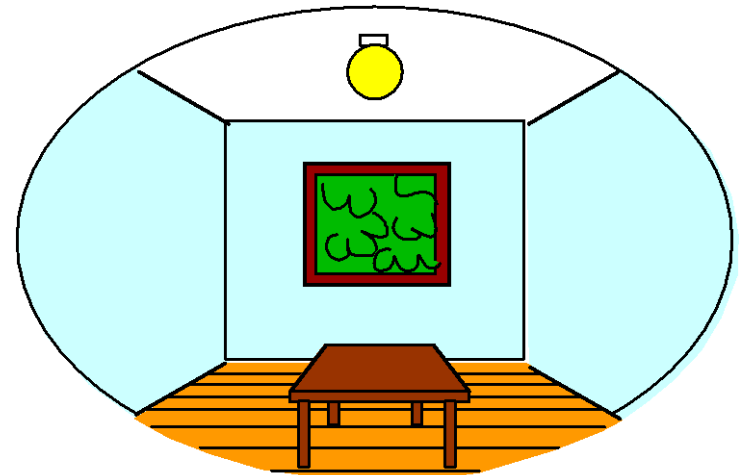
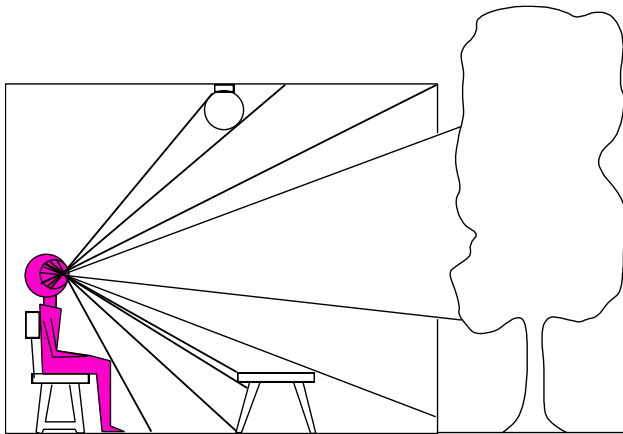
Properties of projection

What is lost?



Fronto-parallel planes

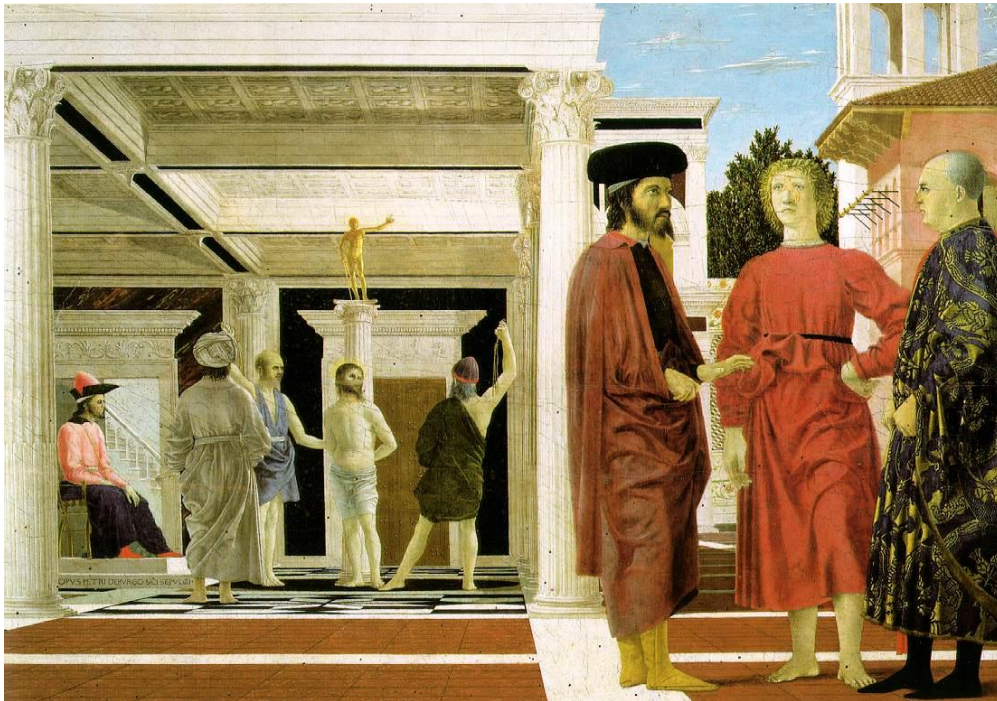
- What happens to the projection of a pattern on a plane parallel to the image plane?
 - All points on that plane are at a fixed *depth* z
 - The pattern gets scaled by a factor of f / z , but angles and ratios of lengths/areas are preserved



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

Fronto-parallel planes

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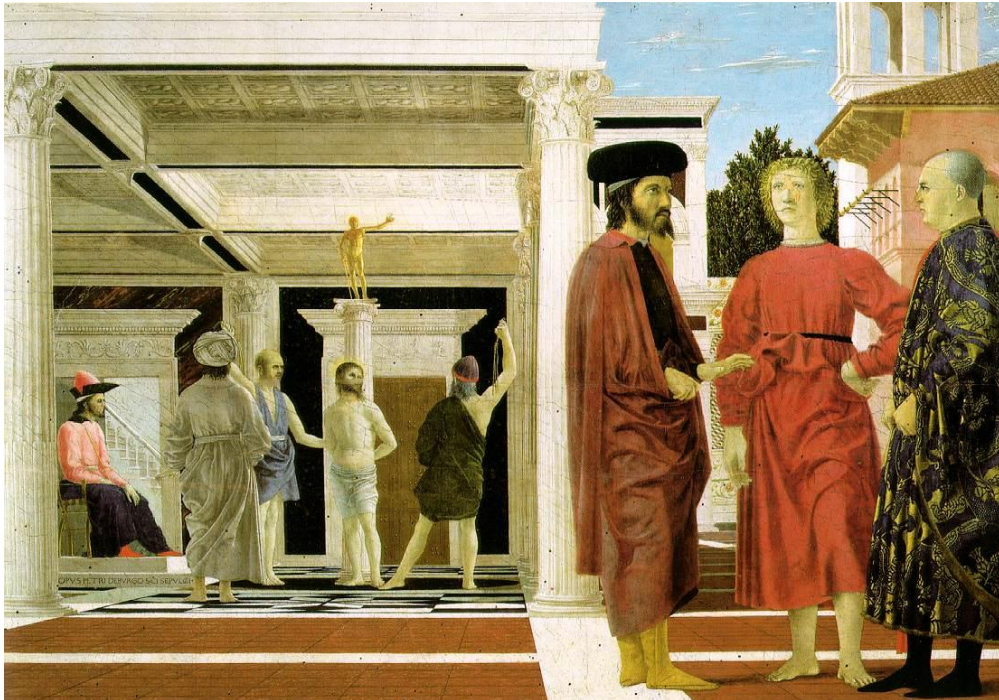


Piero della Francesca, *Flagellation of Christ*, 1455-1460



Jan Vermeer, *The Music Lesson*, 1662-1665

What about non-fronto-parallel planes?



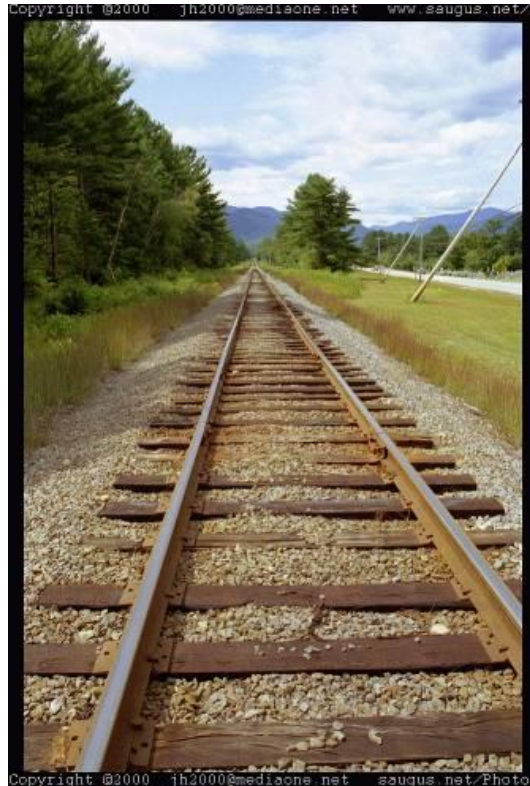
Piero della Francesca, *Flagellation of Christ*, 1455-1460



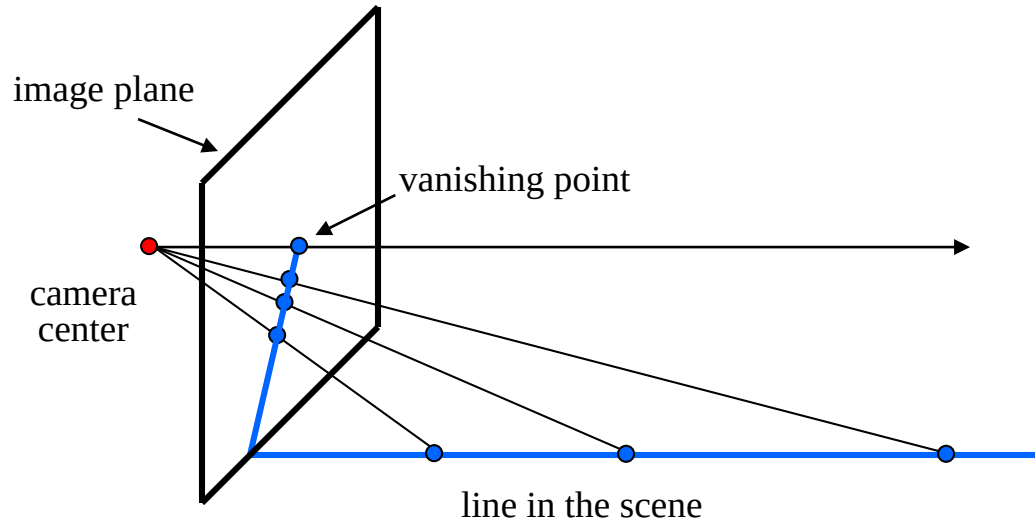
Jan Vermeer, *The Music Lesson*, 1662-1665

Vanishing points

- All parallel lines converge to a *vanishing point*
 - Each direction in space is associated with its own vanishing point
 - Exception: directions parallel to the image plane



Constructing the vanishing point of a line

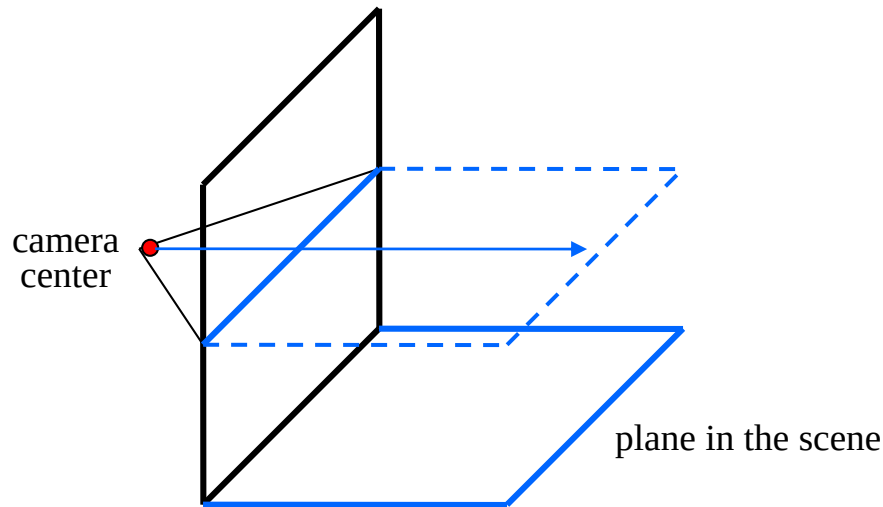
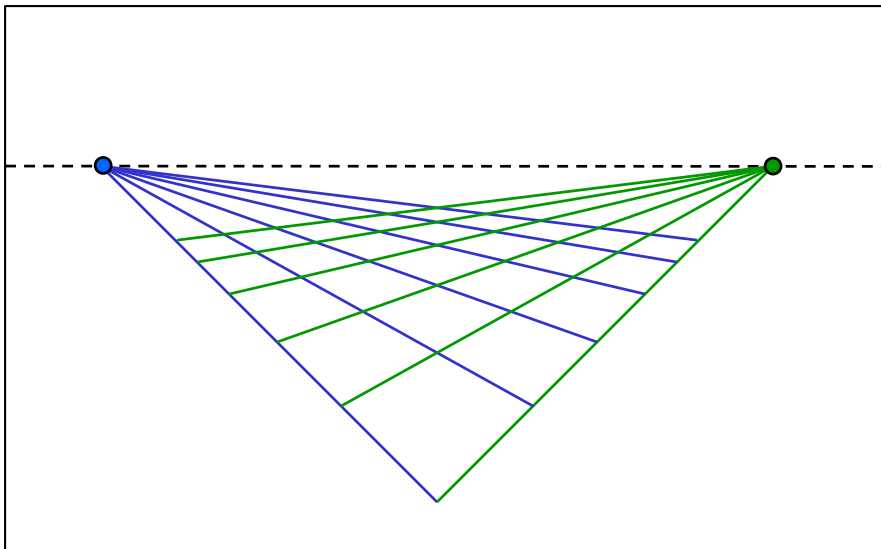


Vanishing lines of planes



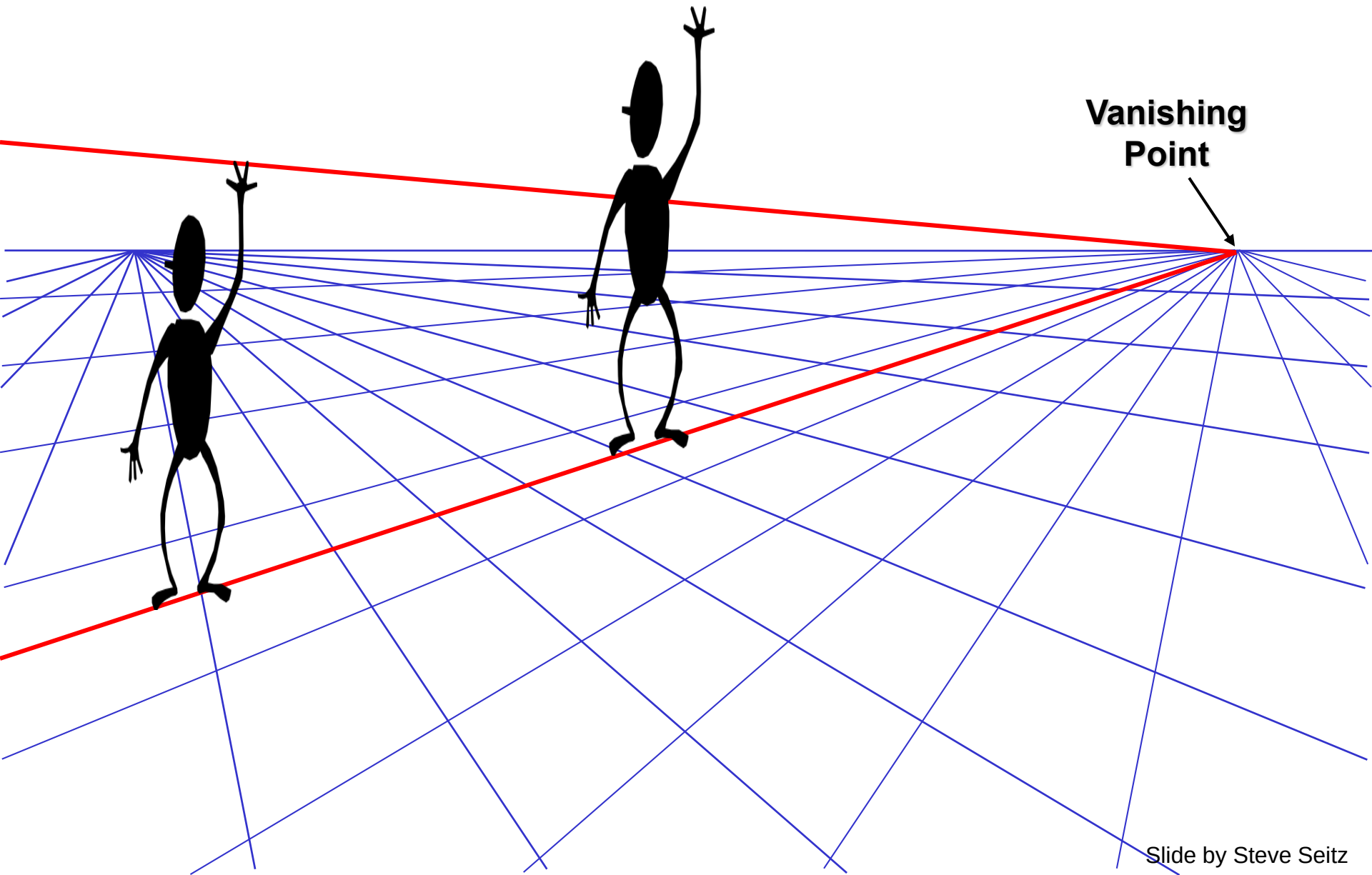
How do we construct the vanishing line of a plane?

Vanishing lines of planes

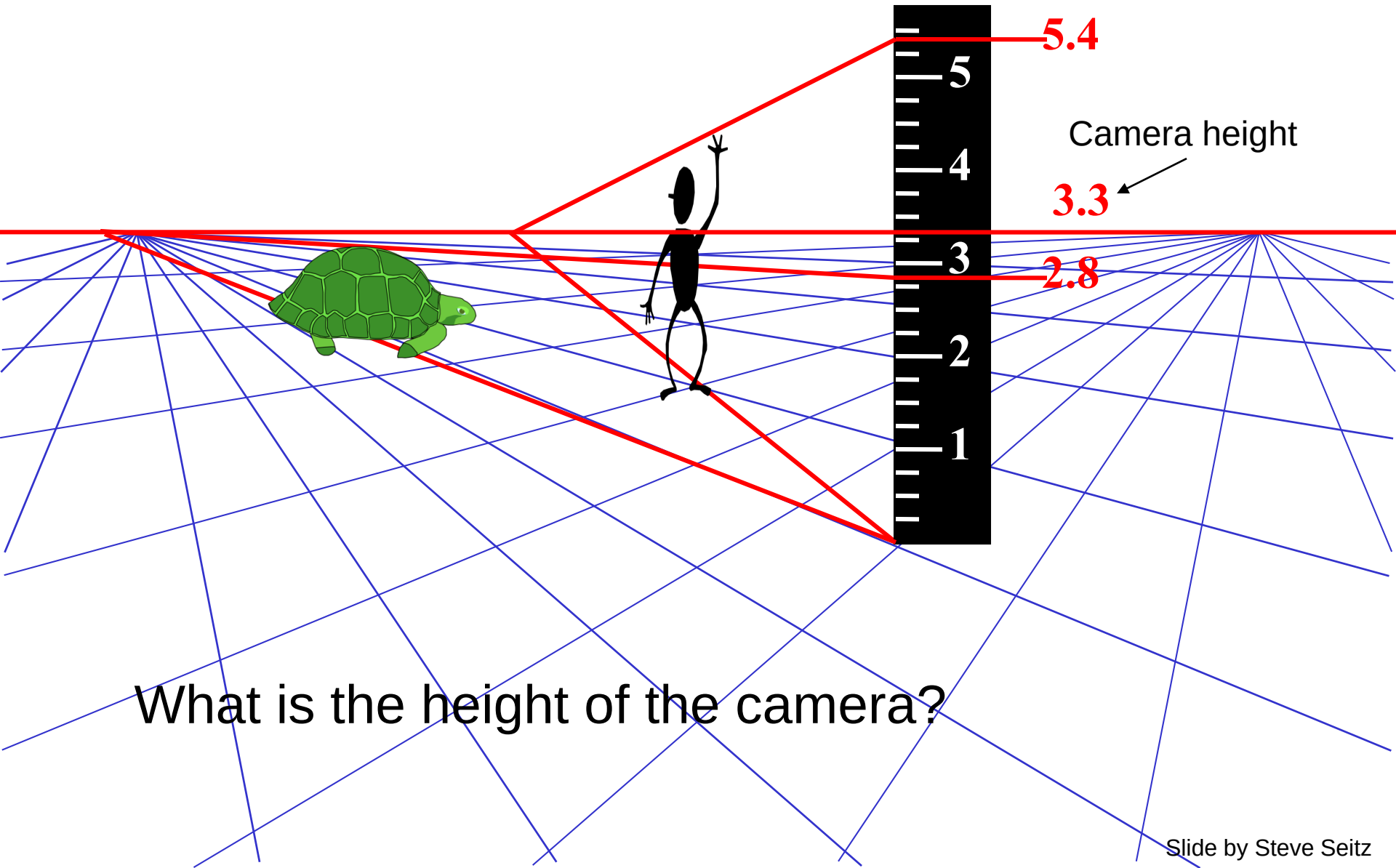


- *Horizon*: vanishing line of the ground plane
 - All points at the same height as the camera project to the horizon
 - Points higher (resp. lower) than the camera project above (resp. below) the horizon
 - Provides way of comparing height of objects

Comparing heights

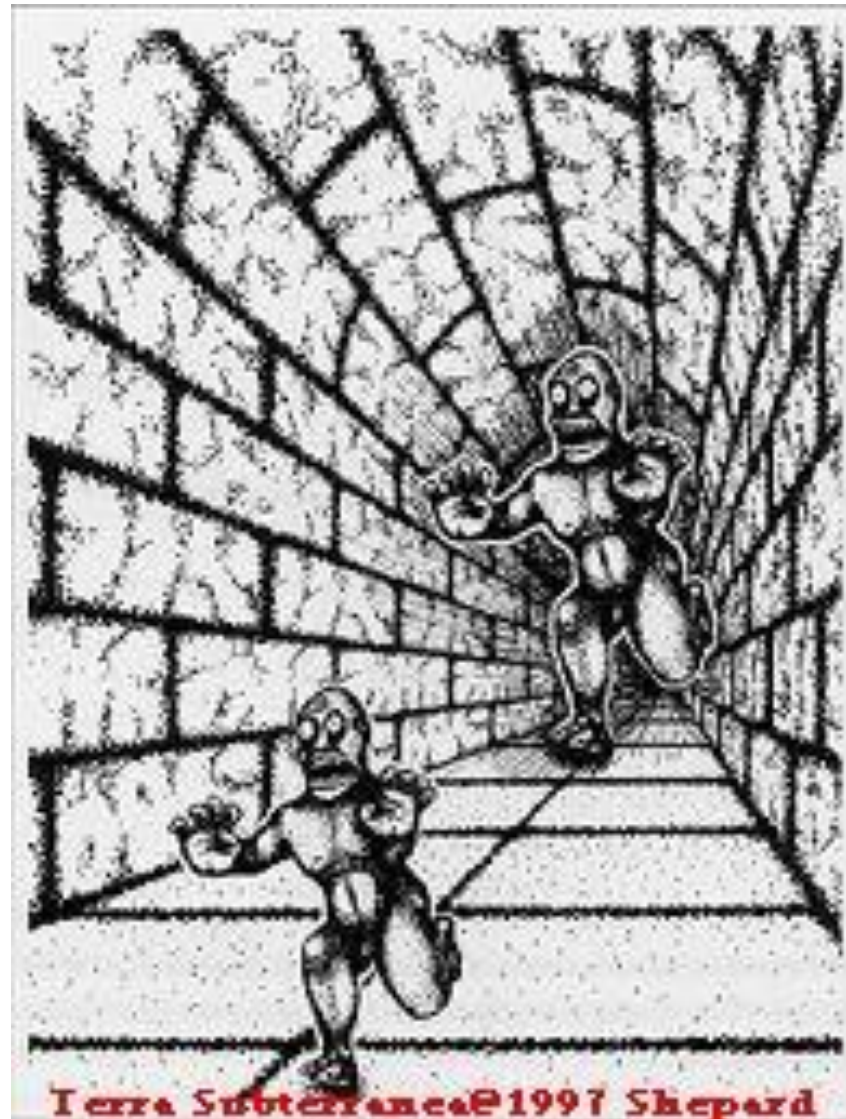


Measuring height

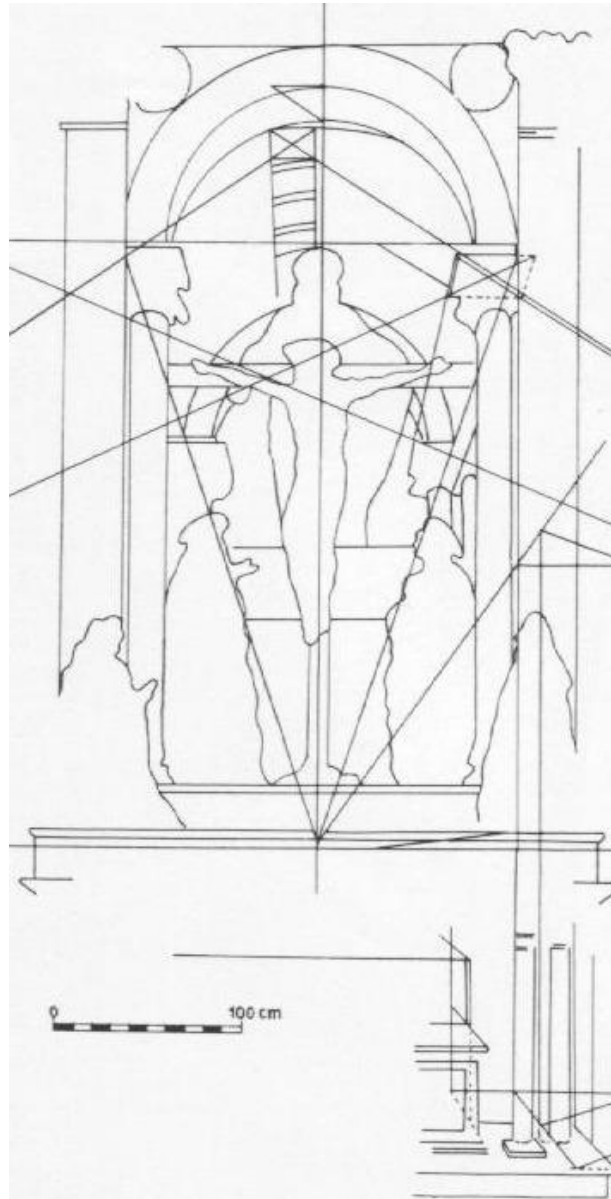
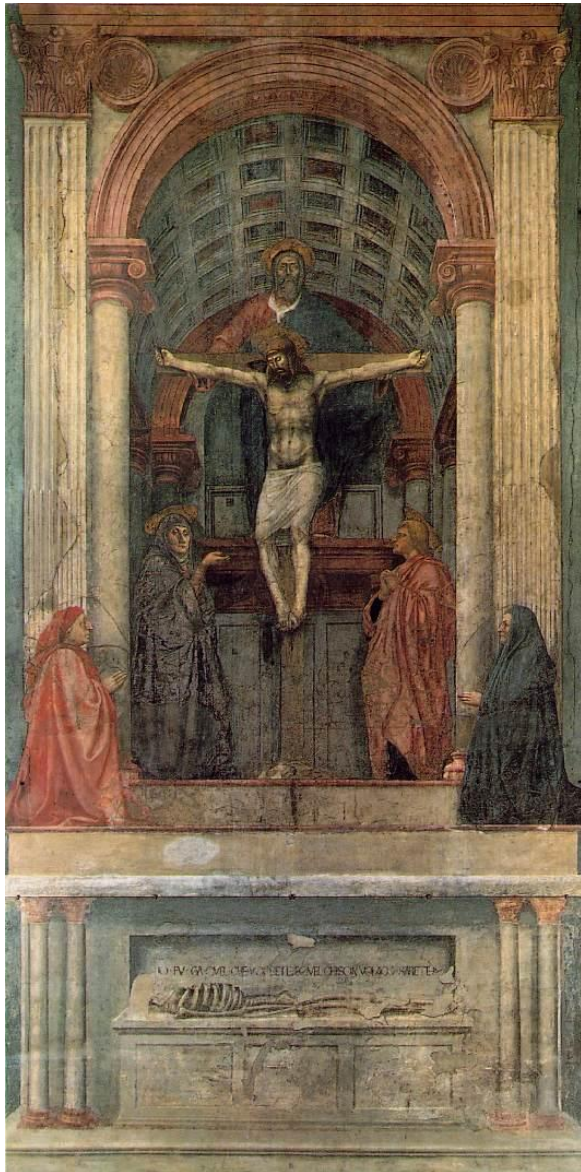


What is the height of the camera?

Perspective cues



Perspective cues in art



Masaccio, *Trinity*, Santa Maria Novella, Florence, 1425-28

One of the first consistent uses of perspective in Western art

Perspective distortion

- What is the shape of the projection of a sphere?

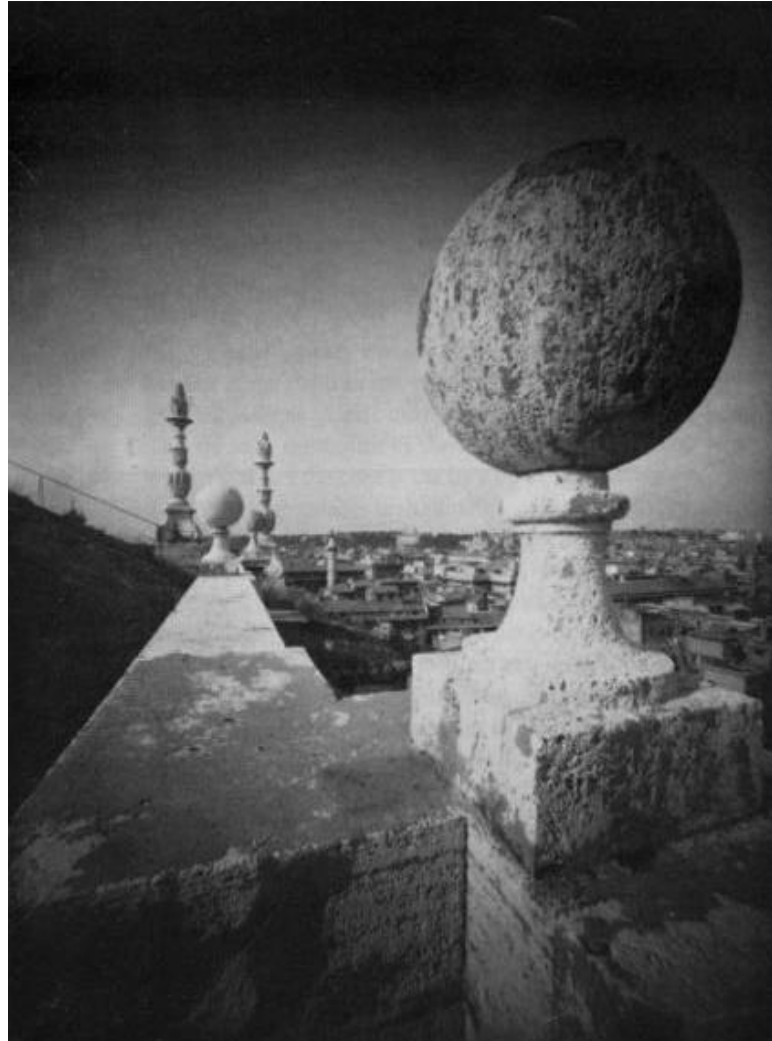
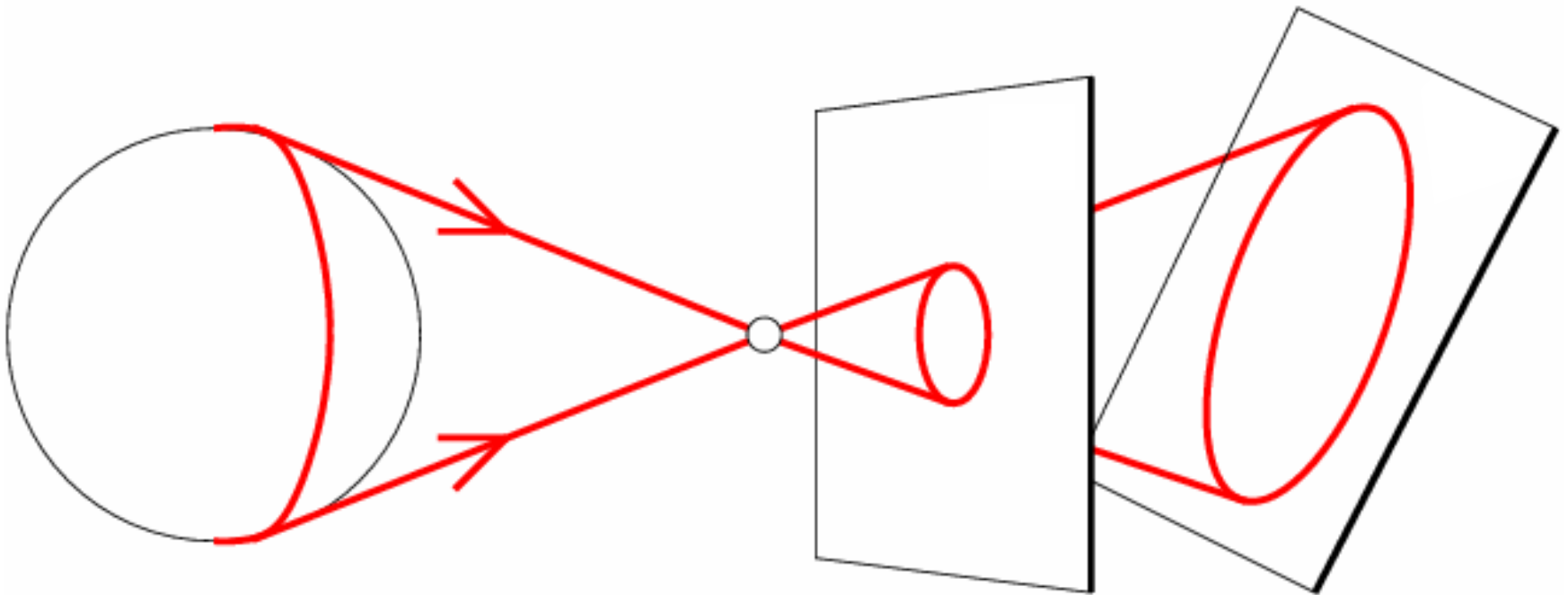


Image source: F. Durand

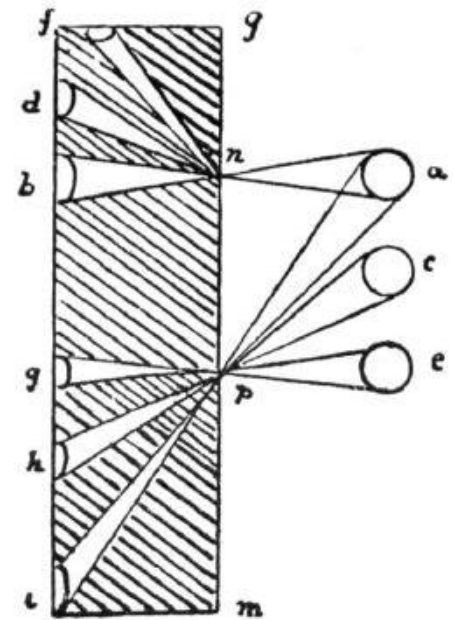
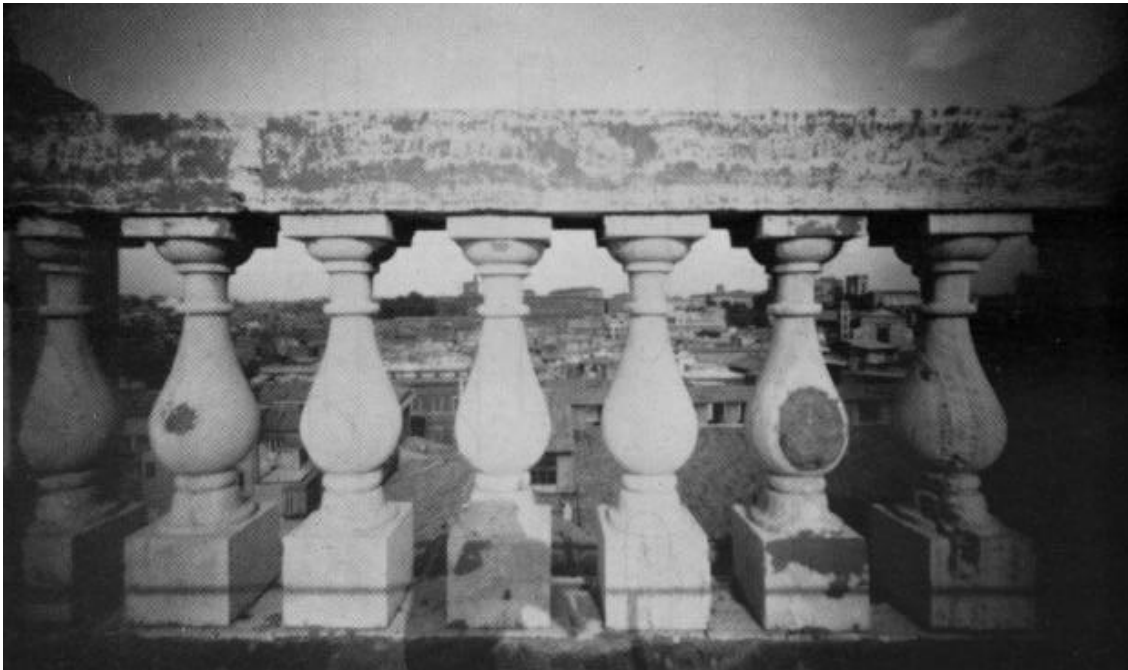
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- What is the shape of the projection of a sphere?



Perspective distortion

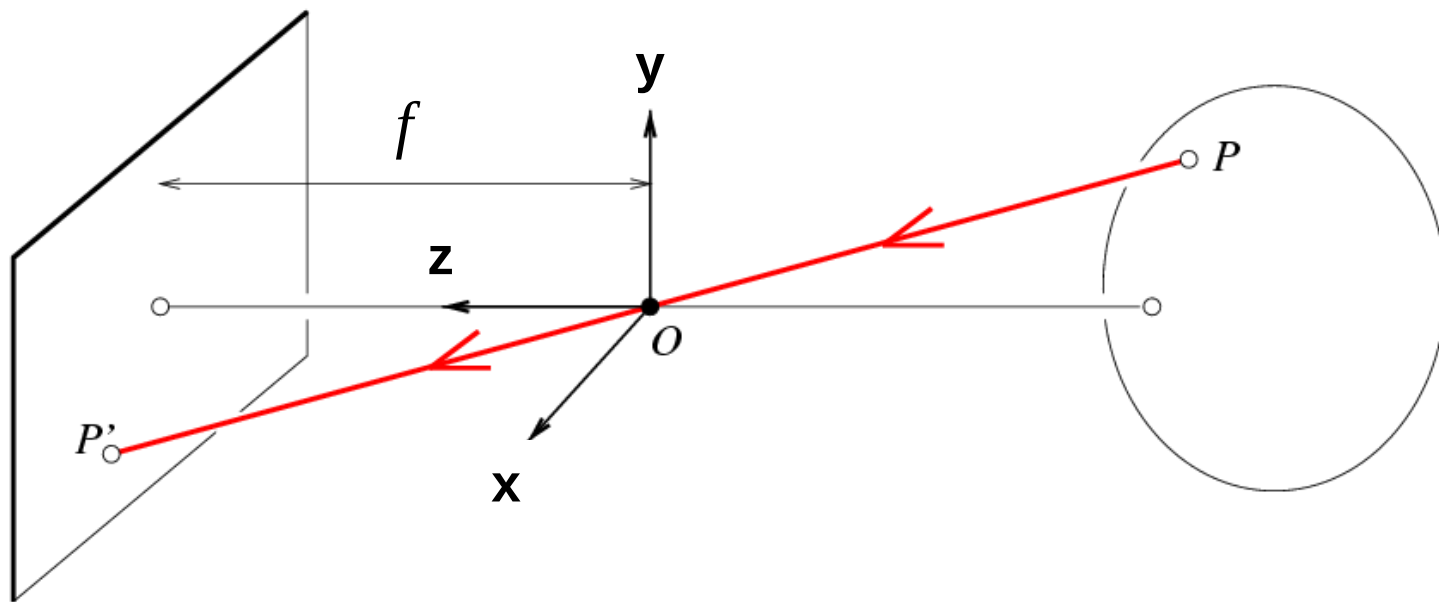
- Are the widths of the projected columns equal?
 - The exterior columns are wider
 - This is not an optical illusion, and is not due to lens flaws
 - Phenomenon pointed out by Da Vinci



Perspective distortion: People



Modeling projection



Projection equation: $(x, y, z) \rightarrow (f \frac{x}{z}, f \frac{y}{z})$

Note: instead of dealing with an image that is upside down, most of the time we will pretend that the image plane is in front of the camera center.

Homogeneous coordinates

$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

Is this a linear transformation?

- no—division by z is nonlinear

Trick: add one more coordinate:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

homogeneous scene
coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \Rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

divide by the third coordinate

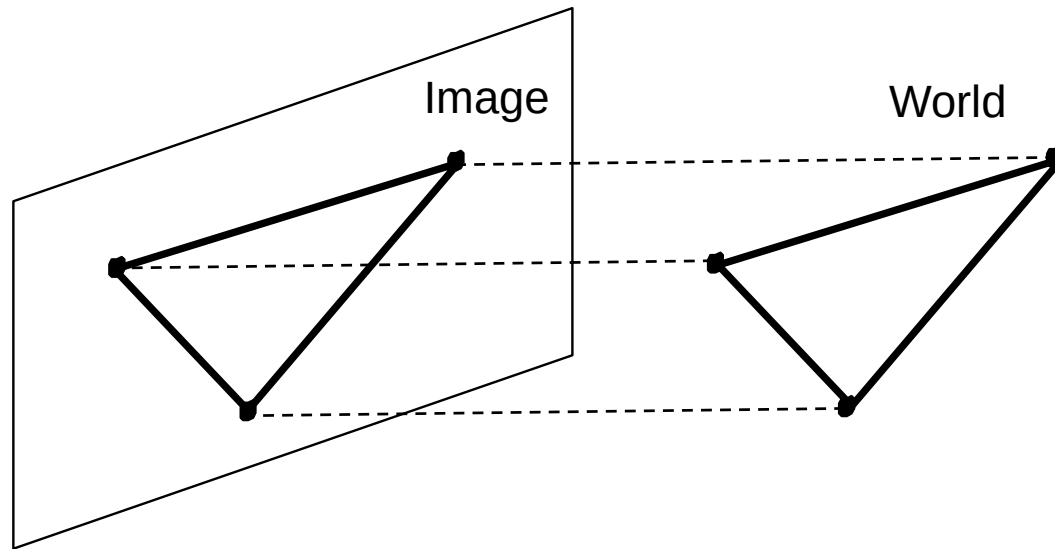
In practice: lots of coordinate transformations...

$$\begin{pmatrix} \text{2D} \\ \text{point} \\ (3 \times 1) \end{pmatrix} = \begin{pmatrix} \text{Camera to} \\ \text{pixel coord.} \\ \text{trans. matrix} \\ (3 \times 3) \end{pmatrix} \begin{pmatrix} \text{Perspective} \\ \text{projection matrix} \\ (3 \times 4) \end{pmatrix} \begin{pmatrix} \text{World to} \\ \text{camera coord.} \\ \text{trans. matrix} \\ (4 \times 4) \end{pmatrix} \begin{pmatrix} \text{3D} \\ \text{point} \\ (4 \times 1) \end{pmatrix}$$

Orthographic Projection

Special case of perspective projection

- Distance from center of projection to image plane is infinite
- Also called “parallel projection”



Orthographic Projection

Special case of perspective projection

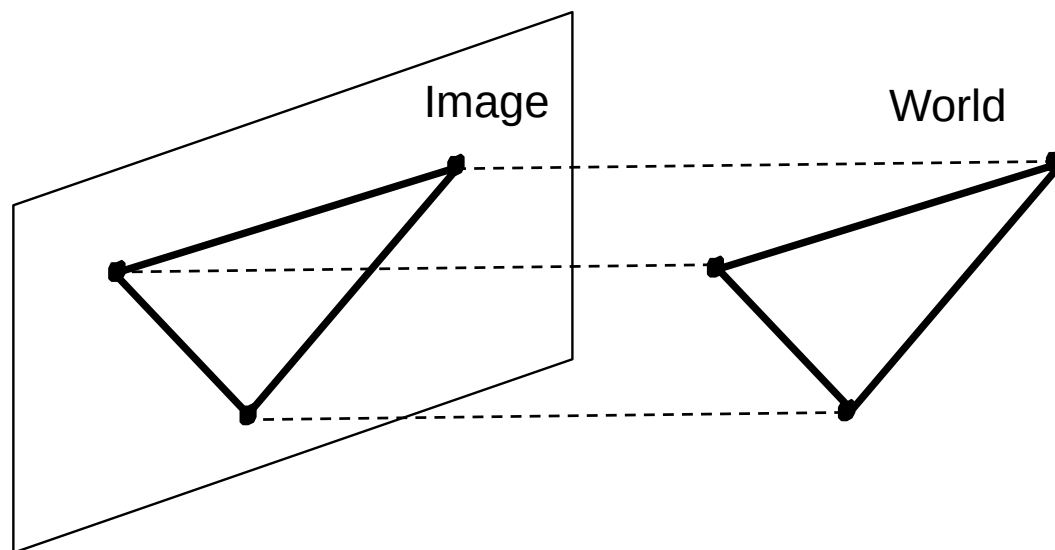
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Orthographic Projection

Special case of perspective projection

- Distance from center of projection to image plane is infinite
- Also called “parallel projection”



- What's the projection matrix?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$